

Dashboard

Project Title: Supply Chain Resiliency & Margin Protection in Prestige Beauty: An Enterprise Integration Case Study

Date: July 31, 2026

Dashboard Overview

This deliverable is an interactive, three-tab Tableau Public dashboard built to accompany the project's datafolio. The dashboard is explanatory in design: each tab is scoped to a specific enterprise stakeholder audience identified in the Project Scoping document (Supply Chain Director, CFO, and Warehouse Operations), and each surfaces the KPIs, filters, and visual encodings most relevant to that audience's decision-making context.

Live Dashboard Link

<https://public.tableau.com/app/profile/shari.nishida/viz/CorrelationOneDataAnalyticsCapstone/ExecutiveSOPRiskMatrix?publish=yes>

Tab 1: Executive S&OP Risk Matrix (Supply Chain Director View)

Surfaces portfolio-wide KPIs (Avg Sephora Rating, Avg Defect Rate, Avg Lead Time, Tariff-Adjusted COGS Inflation) alongside three diagnostic charts: Defect Rate by Product Type, Lead Time Variability by Supplier, and Cost vs. Revenue by SKU under the tariff scenario overlay. Product Type, Supplier, and Transportation Modes filters allow a Supply Chain Director to isolate risk by category or vendor while KPI tiles remain fixed at portfolio-wide totals for consistent benchmarking.

Tab 2: Margin Compression & Simulation (CFO View)

Quantifies the directional gross margin impact of the engineered Tariff Impact Coefficient across product categories via a Margin Waterfall chart and a Baseline vs. Adjusted Margin comparison. A Near-Shore Volume Shift % parameter drives a live Simulated Blended Cost calculation, letting a CFO model the cost impact of shifting procurement volume toward near-shore suppliers. Product Type, Supplier, and Transportation Modes filters apply to the two charts only, so the four KPI tiles remain stable portfolio-wide reference points.

Tab 3: Omnichannel Inventory & WMS Calibration (Warehouse Operations View)

Combines a skincare-scoped Review Volume vs. Stock Posture chart (the underlying Sephora reviews dataset covers skincare exclusively, disclosed directly on the chart), a Reorder Point Table ranking the top SKUs by stockout risk, and a Stockout Risk Scatter with a Critical Stockout Zone reference line at Stock Level = 20. A Product Type filter lets Warehouse Operations isolate reorder priorities and stockout risk by category.

Use Case

A stakeholder opens the dashboard and selects the tab matching their role. Within a tab, they can filter by Product Type, Supplier, or Transportation Mode to narrow the view to their area of concern (e.g., a CFO isolating Skincare's margin compression, or a Warehouse Operations lead isolating Cosmetics stockout risk) while portfolio-wide KPI benchmarks stay visible for comparison. The three tabs are

designed to be read together: Tab 1 identifies where risk concentrates, Tab 2 quantifies what it costs, and Tab 3 translates the findings into an actionable reorder-priority view.

Key Design Decisions & Known Scope Limitations

- A Tariff Sensitivity parameter (0–40%) was built to support dynamic tariff-scenario simulation, but was descoped from the final dashboard in favor of the fixed, category-based Tariff Impact Coefficient (cosmetics 25%, skincare 10%, haircare 0%), which better reflects the documented, category-specific trade-policy basis for this analysis.
- The Sephora reviews dataset (1,094,411 rows) contains skincare reviews only. This is disclosed directly on the datafolio and on Tab 3's review chart, and the associated review-velocity and price-tier growth findings should be read as skincare-scoped, not portfolio-wide.
- Haircare shows \$0.0 margin compression on Tab 2 by design, reflecting its 0% Tariff Impact Coefficient, not a data or calculation error.